

SYSTEM REQUIREMENTS

UNIVERSITY CORRIDOR MEETING ROOM AND WORKSPACE RESERVATION SYSTEM

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TABLE OF CONTENTS

1.	Introduction	3
2.	System Actors.....	4
3.	Functional Requirements	5
4.	Non-functional Requirements / Quality Attributes	7
5.	Security Requirements.....	8
6.	Who Did What?	9
7.	Review checklist.....	9

1. Introduction

<Give an overview of the project here. The overview must highlight the overall objectives of the project and its potential users.>

Overview:

The University Corridor Meeting Room and Workspace Reservation System makes it easier to book meeting rooms and individual workspaces in different university departments. Each department has several corridors, and each corridor contains multiple rooms. Some of these rooms have several workspaces to meet different needs. Workspaces are of different types too. The individual workspace for like studying alone, shared workspace for group studying or office hours. Other rooms may be reserved exclusively for faculty meetings and do not have multiple workspaces; instead, the entire room will be booked for a specified time. The system helps manage these spaces better, reduces booking conflicts, and makes sure the rooms are used efficiently.

Objectives:

- To provide a centralized platform for booking meeting rooms and workspaces.
- To allow users to view, reserve, and manage bookings efficiently.
- To prevent double bookings and enforce booking policies to maintain fair usage.

Potential Users:

- Faculty members for departmental meetings.
- Teaching Assistants (TAs) for office hours and student consultations.
- Senior Year Students for studying alone or in group for project (restricted to 5 members).
- Application system for handling booking, conflict resolution, and notification.
- Admin for creating user accounts based on the user's status.

2. System Actors

<List down the actor names and give a 2-3 lines description of the role of each actor>

Actor Name	Description
Faculty Member	A user who can view, reserve, and manage bookings for both meeting rooms and all types of workspaces.
Teaching Assistant (TA)	A user with the ability to reserve either individual workspaces or shared workspaces for student consultations or meetings (i.e., office hours)
(Senior) Student	A user who can book individual workspaces for study or shared workspace for project work (restricted to 5 members only)
System	The application that handles booking, conflict resolution, and notification.
Admin	The admin is responsible for creating user accounts based on the user's status, such as Faculty Member, Teaching Assistant (TA), or Senior Student.

3. Functional Requirements

<Write system requirements from users' (actors) perspective. Actor names have been highlighted in the sample requirements below. You may group requirements according to actors or modules>

Requirements	
Sr#	Requirement
1	As a faculty member , I want to view available rooms and workspaces so I can book spaces for my meetings.
2	As a faculty member , I want to reserve an entire meeting room or a workspace for a specific time period.
3	As a faculty member , I want to cancel my reservation if I no longer need the room or workspace.
4	As a faculty member , I want to receive notifications about my bookings so that I am kept updated.
6	As a teaching assistant , I want to reserve individual or shared workspaces for student consultations.
7	As a teaching assistant , I want to view the booking history to manage my workspace reservations.
8	As a teaching assistant , I want to receive reminders for my upcoming reservations.
9	As a teaching assistant , I want to search for specific rooms or workspaces based on my teaching schedule.
10	As a senior student , I want to book individual workspaces for studying alone or shared workspaces for group work.
11	As a senior student , I want to view details about available rooms and workspaces to make the best choice.
12	As a senior student , I want to provide feedback and rate the workspaces I have used.
13	As an admin , I want to create user accounts for faculty members, teaching assistants, and senior students.
14	As an admin , I want to manage user accounts and assign roles to ensure proper access control.
15	As a system , I need to handle conflict resolution when multiple users attempt to book the same room or workspace.
16	As a system , I need to automatically approve or reject booking requests based on predefined rules.
17	As a system , I need to send notifications to users regarding booking confirmations, cancellations, and reminders.

18	As a system , I need to generate reports on room and workspace usage for administrators to review.
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4. Non-functional Requirements / Quality Attributes

<Requirements must be testable>

<Security requirements fall in the category of “Non-functional requirements”; however, you need to list them separately in the section **Security Requirements** later in this document.>

Sr#	Requirements
1	The system must respond to user actions (e.g., viewing availability, making reservations) within 2 seconds 95% of the time.
2	The system must support up to 500 concurrent users without any noticeable performance degradation.
3	The system should be available for use 99.9% of the time over a rolling 30-day period.
4	The system should use less than 1 GB of memory during normal operation.

5. Security Requirements

< Go through OWASP top 10 security risks (<https://owasp.org/www-project-top-ten/>).

- (a) Select **3 security risks** that you think are top threats for your system. While doing this, carefully consider the information/functionality that is most vulnerable from security perspective in the context of your project.
- (b) For each security risk (identified above), identify **potential losses** (e.g., financial loss, total business loss, litigation etc.) if you do not take necessary measures to address the above security risks.
- (c) Identify the **controls** (e.g., input validation, audit logs, multi-factor authentication, user roles etc.) that should be implemented in your system in order to address the above security risks.

Sr#	Security Risks	Potential Losses	Controls
1	Injection Attacks (SQL/NoSQL Injection)	Unauthorized access to user data, booking information, and system data integrity compromise.	Implementing parameterized queries and input validation to sanitize all inputs and prevent malicious queries.
2	Broken Authentication	Unauthorized access to the system, leading to data breaches and unauthorized booking or cancellation actions.	Implementing secure password storage (hashing and salting), enforcing strong password policies.
3	Cross-Site Scripting (XSS)	Data theft, session hijacking, and spread of malicious scripts.	Validating and sanitizing all user inputs, and escaping special characters in input fields.

6. Who Did What?

Name of the Team Member	Tasks done
Syed Meesum Raza Naqvi	Completed the following sections: Software Process Selection, Functional Requirements, Security Requirements, Gantt Chart, Development Environment Preparation, Deployment Platform, and Review Checklist.

7. Review checklist

Before submission of this deliverable, the team must perform an internal review. Each team member will review one or more sections of the deliverable.

Section Title	Reviewer Name(s)
Introduction	Syed Meesum Raza Naqvi
Actors	Syed Meesum Raza Naqvi
Functional Requirements	Syed Meesum Raza Naqvi
Non-functional requirements	Syed Meesum Raza Naqvi
Security Requirements	Syed Meesum Raza Naqvi